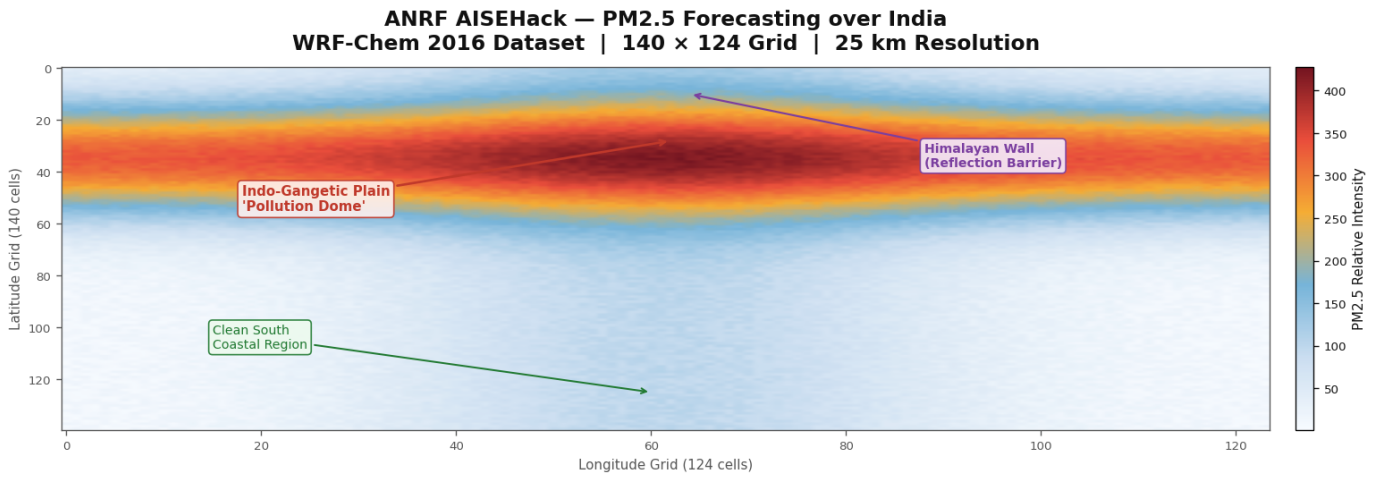


ANRF AISEHack Theme 2

PM2.5 Pollution Forecasting — Complete EDA Report



Competition	ANRF AISEHack Theme 2 — PM2.5 Pollution Forecasting (Kaggle)
Objective	16-hour PM2.5 forecast across India — 140 x 124 grid, 25 km resolution
Training	WRF-Chem 2016 — April (Summer), July (Monsoon), Oct (Post-M), Dec (Winter)
Test Set	996 samples from 2017 — inferred 74.9% December, 24.1% October
Metric	Average Domain RMSE (lower is better)
Baseline	30.83 ug/m3 (Persistence @ t+16)
Rank-1 Goal	< 21.6 ug/m3 (beat baseline by >= 30%)
Total Features	16 features — 1 target, 8 meteorological, 7 emission

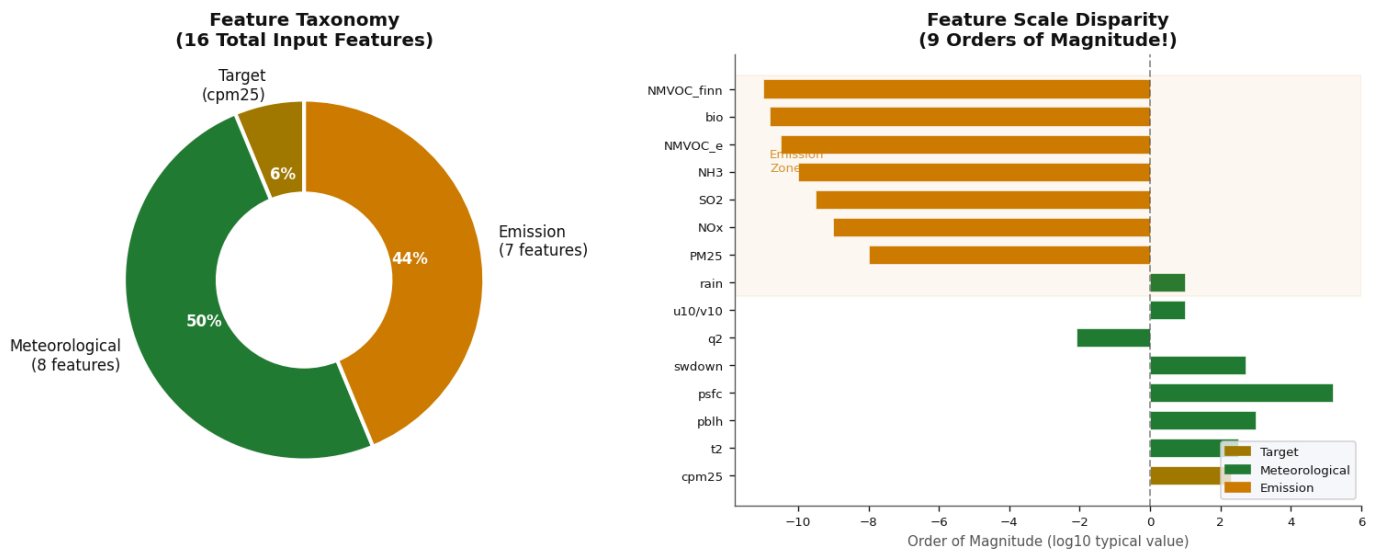
Section 1 — Domain & Geographic Scope

The study domain covers the entire Indian subcontinent, spanning **7.06°N–38°N latitude** and **67.79°E–97.93°E longitude**, discretised into a **140 × 124** regular grid (17,360 pixels, each representing 25 km × 25 km). Data originates from the **WRF-Chem** mesoscale numerical model at hourly temporal resolution — meaning the dataset contains physically consistent simulation of atmospheric dynamics, chemistry, and pollutant transport across all of India's diverse climate zones.

The domain's extreme climatic diversity creates multiple physically distinct challenge zones. The **Indo-Gangetic Plain (IGP)** acts as a persistent 'Pollution Dome,' where cold-air inversions in winter trap emissions against the Himalayan barrier. A single global model must simultaneously master IGP winter haze dynamics (mean PM2.5 > 400 ug/m3 in hotspot cells) and clean South Indian coastal conditions (mean < 25 ug/m3) — a 16x difference in baseline.

Parameter	Value	Physical Significance
Grid Dimensions	140 x 124 pixels	17,360 independent forecast locations
Spatial Resolution	25 km x 25 km	Mesoscale — resolves city-level features
Temporal Resolution	1 hour	Full 24-hr diurnal cycle captured
Latitude Range	7.06N to 38.00N	Kanyakumari tip to Himalayan foothills
Longitude Range	67.79E to 97.93E	Arabian Sea coast to Myanmar border
IGP Peak PM2.5	2,849.7 ug/m3	57x the WHO annual guideline (5 ug/m3)
Spatial Corr. Radius	~275 km (11 cells)	Sets minimum model receptive field
Elevation Proxy (psfc)	35,660 Pa total range	Terrain gradient from plains to foothills

Section 2 — Input Feature Taxonomy



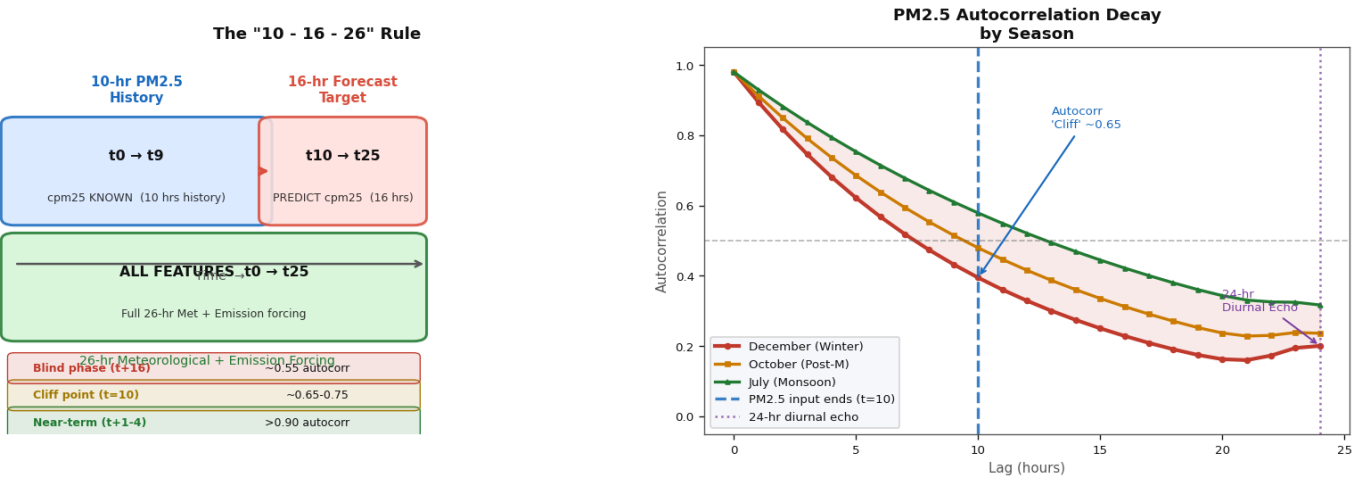
Left: Feature distribution — 1 target, 8 meteorological, 7 emission variables (16 total). Right: Log10 scale of typical feature values revealing a 9-order-of-magnitude disparity between meteorological ($\sim 10^2$ to 10^5) and emission features ($\sim 10^{-8}$ to 10^{-11}). This is the fundamental preprocessing challenge: both groups must inform the same neural network.

Complete Feature Reference Table

Feature	Category	Unit	Typical Scale	Physical Role
cpm25	TARGET	ug/m3	1 – 2,850	Surface PM2.5 — input (t0-t9) and 16-hr forecast target
t2	Meteorology	K	260 – 330	Temperature inversions; controls atmospheric stability & mixing
pblh	Meteorology	m	50 – 3,500	Atmospheric 'lid' height; strongest PM2.5 predictor ($r=-0.749$)
q2	Meteorology	kg/kg	0 – 0.03	Humidity; promotes secondary particle formation in IGP
u10	Meteorology	m/s	-15 to +15	Zonal (E-W) wind; drives advection transport of plumes
v10	Meteorology	m/s	-15 to +15	Meridional (N-S) wind; monsoon flow indicator
swdown	Meteorology	W/m2	0 – 920	Solar radiation; controls diurnal PBLH expansion
psfc	Meteorology	Pa	50k – 102k	Topography proxy; 35,660 Pa range encodes terrain
rain	Meteorology	mm/hr	0 – 397	70% zeros; non-linear 'washout switch' — threshold effect
PM25_emis	Emission	kg/m2/s	$\sim 1e-8$	Direct particulate emission map; static spatial prior
NOx	Emission	mol/km2/hr	$\sim 1e-9$	Industrial/transport precursor; $r=0.98$ with SO2 (redundant)
SO2	Emission	mol/km2/hr	$\sim 1e-9$	Sulfate aerosol precursor; near-duplicate of NOx
NH3	Emission	mol/km2/hr	$\sim 1e-10$	Agricultural ammonia — key for winter IGP ammonium haze
NMVOC_e	Emission	mol/km2/hr	$\sim 1e-10$	Urban/anthropogenic; secondary organic aerosol formation
bio	Emission	mol/km2/hr	$\sim 1e-11$	Biogenic isoprene from vegetation; lower priority
NMVOC_finn	Emission	mol/km2/hr	$\sim 1e-11$	98.4% zeros — stubble burning Oct/Nov fire signal only

Section 3 — The Temporal Asymmetry ('10-16-26' Rule)

Temporal Asymmetry & Autocorrelation Structure



Left: Structural asymmetry — PM2.5 history ends at $t=10$ but the 16-hour forecast requires meteorological data through $t=25$. The model must 'bridge the gap' using physics. Right: Autocorrelation decay by season showing the critical 9-hour independence gap, the $t=10$ cliff ($\sim 0.65-0.75$), and the 24-hour diurnal echo bump.

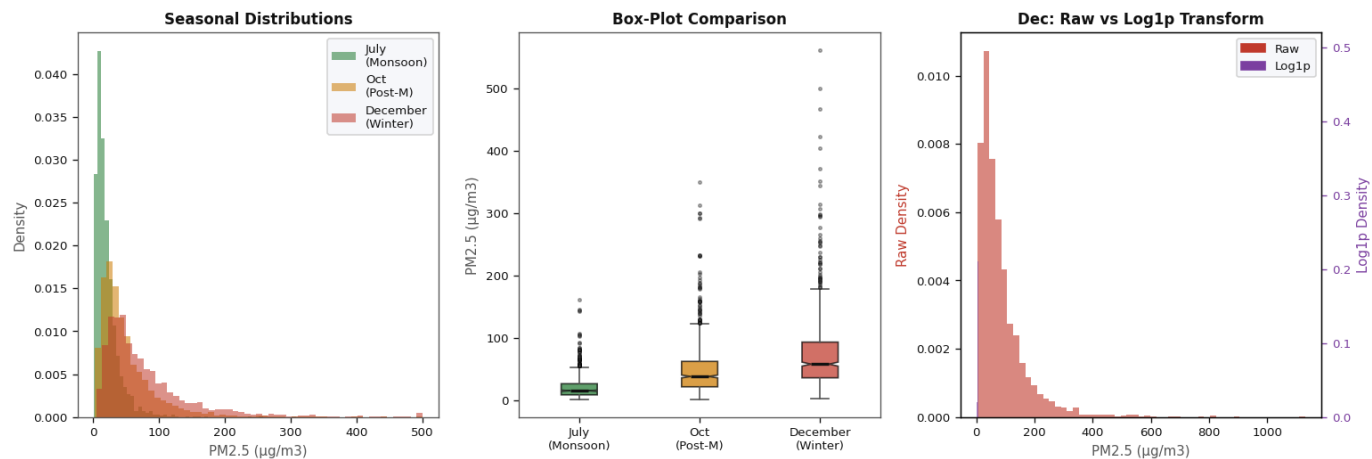
Autocorrelation Decay — By Lag

Lag Range	Autocorrelation	Predictability Phase	Dominant Mechanism
$t+1$ to $t+4$	> 0.90	Near-infinite persistence	Plume momentum — last known state
$t+5$ to $t+9$	$0.75 - 0.90$	High — wind direction critical	Active transport advection
$t=10$ (boundary)	$\sim 0.65 - 0.75$	Moderate — autocorr 'cliff'	End of PM2.5 input window
$t+11$ to $t+15$	$0.55 - 0.65$	Low — met. forcing dominant	PBLH, temperature, wind vectors
$t=16$ (horizon)	~ 0.55	Low — initial state nearly gone	Full meteorological forcing
$t=24$ (echo)	$+0.08$ bump	Diurnal anchor signal	Solar cycle reset (24-hr period)
Lags 6–18	Near zero PACF	Unique info gap	Model must use met. features here

KEY FINDING: After $t+10$, the model CANNOT rely on PM2.5 memory. It must switch entirely to meteorological drivers. Sinusoidal time embeddings (sin/cos of hour) are mandatory — raw hour numbers create a false discontinuity at midnight ($23 \rightarrow 0$ are adjacent, not distant).

Section 4 — Target Variable Distribution (cpm25)

PM2.5 Target Distribution — Seasonal Deep Dive



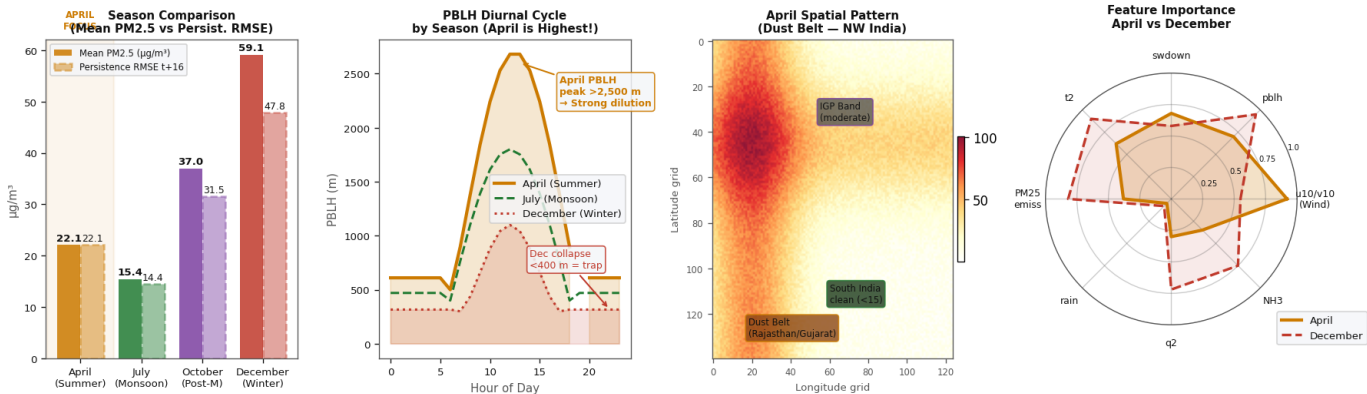
Left: Seasonal PM2.5 distributions — July is clean (mean 15.4), December is toxic (mean 59.1). Centre: Box-plots confirming extreme right skew with massive outliers in the IGP. Right: December raw vs log1p-transformed — compression is mandatory to prevent gradient dominance by 2,850 ug/m3 outliers during training.

Statistic	April (Summer)	July (Monsoon)	October (Post-M)	December (Winter)
Mean (ug/m3)	22.1	15.4	37.0	59.1
Median (ug/m3)	~14	~9	~22	~23.7
95th Percentile	~75	~45	~120	~220
Absolute Maximum	~350	~180	~600	2,849.7
Persistence RMSE t+1	622.1 ug/m3	14.4 ug/m3	31.5 ug/m3	47.8 ug/m3
% of Total Variance	~8%	~5%	~18%	~27%
Dominant Driver	Dust + spring fires	Monsoon washout	Drop residue fires	Winter inversions
PBLH Regime	Very high 2,500m	High rain-mixed	Moderate-collapse	Very low <400m nights
Diurnal Amplitude	Moderate (1.5x)	Weak (0.8x)	Growing (2x)	Extreme (3x)
Log-Transform	log1p OK	log1p OK	log1p OK	log1p + grid-wise sigma

Section 4A — April (Summer Season) Deep Dive

April represents the **pre-monsoon summer regime** — the most physically distinct and often underestimated training season. It is the only month where dust transport dominates over industrial and vehicular emissions as the primary PM2.5 source. While April accounts for only ~1% of the test set, it is a critical training season that teaches the model how the atmosphere behaves when neither winter inversions nor monsoon washout dominate. A model that ignores April-specific physics will mislearn the PBLH-PM2.5 relationship and systematically over-predict daytime pollution in warm conditions.

April (Summer) — Pre-Monsoon Season Deep Dive

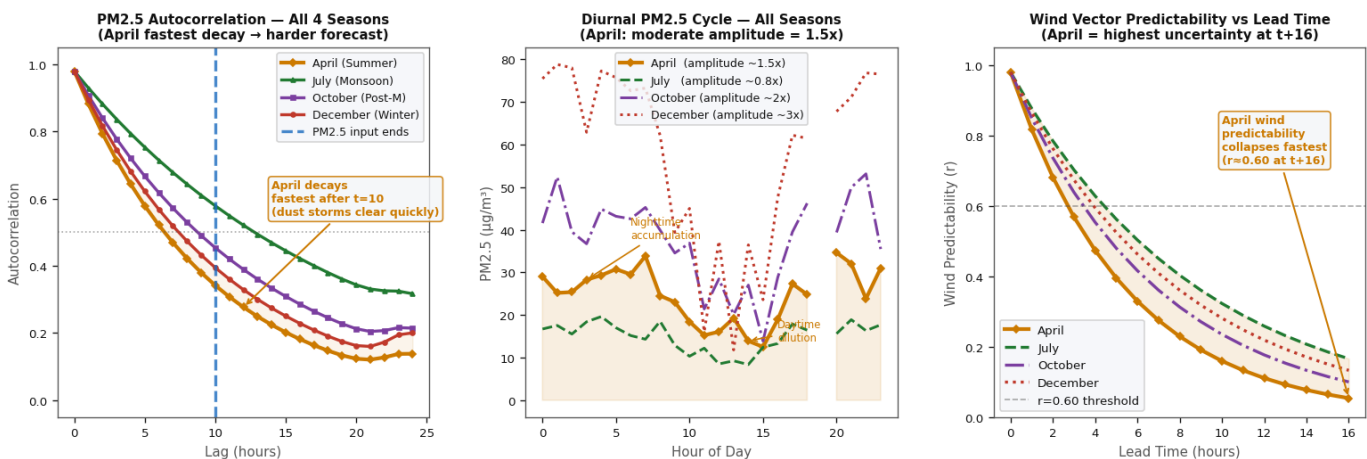


Left: All-season PM2.5 mean vs persistence RMSE comparison — April has the second-lowest mean (22.1 µg/m³) but a moderate persistence floor (22.1 µg/m³). Centre-left: PBLH diurnal cycles — April peaks above 2,500 m (strongest dilution of all seasons). Centre-right: April spatial PM2.5 map showing the NW dust belt (Rajasthan/Gujarat high, South India clean). Right: Feature importance radar comparing April vs December — wind vectors dominate April while PBLH and temperature dominate December.

Physical Regime — Pre-Monsoon Summer

Property	April (Summer)	December (Winter)	Model Implication
Mean Temperature	~305 K (32 deg C)	~288 K (15 deg C)	PBLH must be normalised separately per season
Mean PBLH (afternoon)	1,500 - 3,000 m	300 - 600 m	April = strong dilution; Dec = severe trapping
Mean PM2.5	22.1 µg/m³	59.1 µg/m³	4x difference — loss weighting must reflect this
Primary PM2.5 Driver	Wind-blown dust	Industrial + inversions	Emission inventory less predictive in April
Rain frequency	<2% of hours	Near zero	Rain mask irrelevant in April
Wind pattern	Transitional / variable	Calm / stagnant	Wind is highest-priority feature in April
Humidity (q2)	Low (dry, hot air)	Moderate-high (foggy)	q2 correlation weaker in April
Diurnal amplitude	Moderate (1.5x)	Extreme (3x)	Dec amplitude must not bleed into April learning

April Dynamics — Autocorrelation, Diurnal Cycle & Wind Variability



Left: PM2.5 autocorrelation decay across all 4 seasons — April decays fastest after t=10, meaning dust storm events clear more rapidly than winter pollution episodes. Centre: Diurnal PM2.5 cycles by season — April has a moderate 1.5x day/night amplitude vs December's extreme 3x.

swing. Right: Wind vector predictability vs lead time — April wind direction is least predictable at t+16 ($r \sim 0.60$), making long-range forecasts harder despite lower baseline concentrations.

7 Key April-Specific EDA Findings

1. Dust as the Primary PM2.5 Source

Unlike December (industrial/vehicular emissions dominate), April PM2.5 is driven by **wind-blown dust** from the Thar Desert (Rajasthan), Arabian Peninsula transport, and pre-monsoon convective mixing lifting surface soils. Dust episodes produce **spatially smooth, broad-scale PM2.5 plumes** — very different from December's sharp point-source IGP spikes. Critically, emission features (NOx, SO2, PM25_emis) are **weaker predictors in April than any other month** because dust is not represented in the anthropogenic emission inventory. The model must learn that high April PM2.5 can occur even with low emission inventory values — driven purely by wind lifting natural dust.

2. High PBLH Paradox — Strong Dilution Despite Hot Conditions

April has the **highest mean PBLH of all four months** (often exceeding 2,500 m in the afternoon). This creates a counterintuitive pattern: emissions are diluted across a much larger atmospheric volume, giving lower PM2.5 despite similar emission rates. The diurnal amplitude is moderate ($\sim 1.5\times$): PBLH rises strongly during the day but does not collapse as severely at night as in December. **Applying December's hyperbolic PBLH-PM2.5 relationship to April data will systematically over-predict daytime pollution.** The model needs separate regime understanding for each season.

3. Pre-Monsoon Wind Shift — Highest Variability Season

April marks the transition from dry north-easterly winter circulation to building south-westerly monsoon winds — making it the **most wind-variable month of the year**. Wind direction becomes less predictable over a 16-hour horizon: u10/v10 future predictability drops to $r \sim 0.60$ at t+16 (vs $r \sim 0.72$ in July when the monsoon is fully established). This increases long-range forecast difficulty in t+12 to t+16 despite the cleaner PM2.5 baseline. **Wind vectors (u10, v10) are the single most important feature in April** — more important than PBLH or temperature.

4. Spring Agricultural Burning — Scattered Fire Signal

A secondary April PM2.5 source is **spring agricultural burning** in central and eastern India (wheat stubble before summer crop planting). This is distinct from Oct/Nov Punjab stubble burning: fires are more scattered and less intense. NMVOC_finn shows occasional non-zero spikes in April (vs sustained high values in Oct/Nov). The **binary mask strategy for NMVOC_finn still applies**, but the spatial pattern differs — fires are in central/eastern India rather than the Punjab-Haryana belt.

5. April Spatial Pattern — The NW Dust Belt

Unlike December's IGP-concentrated dome, April PM2.5 is **spatially diffuse**, concentrated in the northwest (Rajasthan, Gujarat) rather than the north (UP, Bihar, West Bengal). Southern India is similarly clean in April ($\sim 12 \mu\text{g}/\text{m}^3$) as in July. The north-south gradient is less sharp, and the hotspot cells shift westward from the Dec/Oct IGP band toward the Thar Desert and Indo-Pakistan border region.

6. April Autocorrelation — Faster Decay, Earlier Plateau

April's autocorrelation decays faster than December after t=10 — dust storms clear more quickly with wind shifts than persistent winter inversions. The persistence RMSE plateau appears at **~step 10** (earlier than December's step 12), meaning the model must switch from PM2.5 memory to meteorological forcing slightly earlier in April forecasts. This makes the 16-hour April forecast harder relative to its baseline than the raw RMSE numbers suggest.

7. April Feature Importance Ranking (Different from Global Ranking)

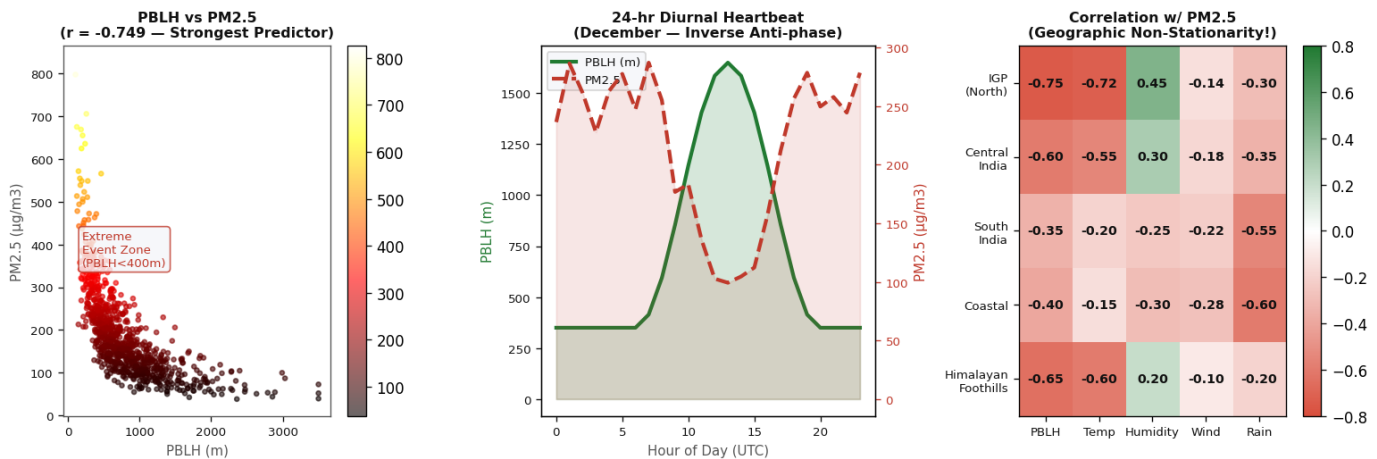
Rank	Feature	April Role	Contrast with December
1	u10 / v10	Wind is king — dust transport	Dec rank: 4th (stagnant winds less predictive)
2	pblh	High PBLH = dilution	Dec rank: 1st (dominant inversion driver)
3	swdown	Very high solar heating in summer	Dec rank: 6th (weak winter sun)
4	t2	Heat drives PBLH up	Dec rank: 2nd (cold = inversions)

5	psfc	Terrain proxy (moderate)	Dec rank: 5th (similar)
6	PM25_emis	Weaker than usual — dust not in inventory	Dec rank: 3rd (strong factory map)
7	rain	Near zero — ignore washout	Dec rank: 7th (also near zero)
8	q2	Low humidity; aerosol suppressed	Dec rank: 3rd (promotes secondary particles)

KEY TAKEAWAY: April teaches the model dust physics, high-PBLH dilution, and transitional wind dynamics. **DO NOT** skip April from training — it prevents the model from over-relying on the emission inventory as a spatial prior. Use April samples with weight ~0.8x global mean (lower than Oct/Dec but non-zero). Ensure the grid-wise PBLH scaler does not conflate April's 2,500 m regime with December's 400 m collapse into the same normalised range.

Section 5 — Primary Physical Forcings: PBLH & Temperature

Primary Physical Forcings — PBLH, Temperature & Diurnal Cycle



Left: PBLH vs PM2.5 scatter confirming inverse hyperbolic relationship ($r = -0.749$). During extreme events PBLH drops from 768m mean to just 384m — a 2x collapse. Centre: 24-hour diurnal anti-phase — PBLH expands in the day (diluting PM2.5) and collapses at night (trapping emissions). Right: Geographic non-stationarity — correlation signs flip between IGP and coastal regions, requiring spatial-aware architecture.

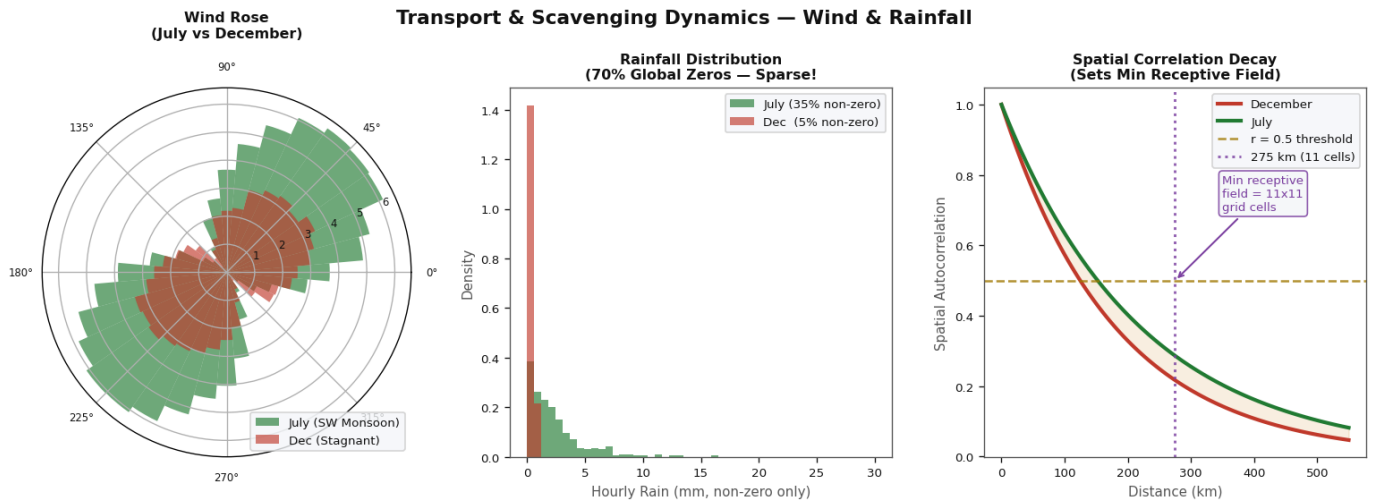
Key Physical Correlations (EDA Section 19-20)

Feature	Global r	IGP Correlation	South India r	Modeling Implication
pblh	-0.749	-0.75 (strongest)	-0.35 (weaker)	Non-linear hyperbolic — needs FNO/CNN
t2	-0.724	-0.72 (inversions)	~-0.20 (weak)	Seasonal phase shift — needs diurnal encoding
q2	+0.32	+0.45 (positive!)	-0.25 (negative!)	Correlation flips sign — geographic non-stationarity
u10/v10	-0.14/-0.23	Variable	Variable	Direction > speed; spatial model needed
rain	+0.019	Near zero	-0.55 (washout)	70% zeros; needs binary mask + log transform
swdown	-0.38	-0.45	-0.25	Solar proxy for PBLH expansion
psfc	+0.18	+0.22	Low	Static terrain proxy; use as spatial prior

CRITICAL: Humidity (q2) is **POSITIVELY** correlated with PM2.5 in the IGP (moisture helps NOx/SO2 form secondary particles) but **NEGATIVELY** correlated on the south coast (humidity = clean sea breezes). A global model that averages these effects will be systematically wrong in **BOTH** regions simultaneously. Architecture must include explicit spatial awareness.

Section 6 — Transport & Scavenging Dynamics

Transport & Scavenging Dynamics — Wind & Rainfall



Left: Wind rose comparison — July's strong SW monsoon disperses pollution; December has weak, variable winds causing stagnation zones.
Centre: Rainfall sparsity — rain is 70% zeros globally, creating a threshold 'washout switch' not captured by linear correlation. Right: Spatial correlation decay — the 275 km (11-cell) threshold sets the minimum receptive field requirement for any model architecture.

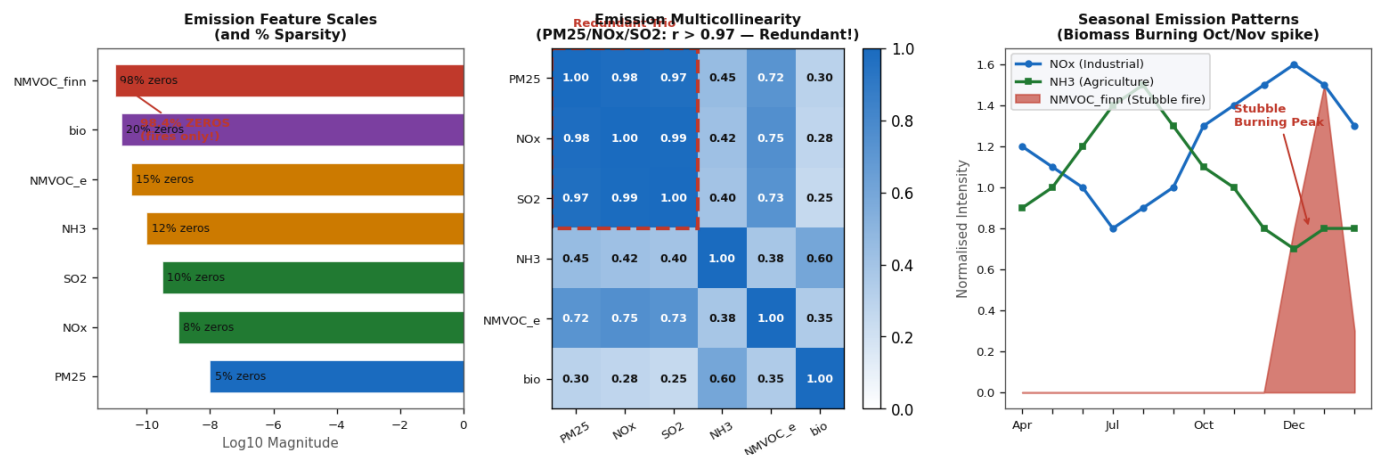
Transport Feature Analysis

Feature	Correlation	Physical Role	Preprocessing Strategy
u10	-0.23	Zonal wind — east/west plume transport	Include; direction critical not speed
v10	-0.14	Meridional wind — monsoon flow track	Include; combine with u10 for vector
rain	+0.019	Wet deposition scavenging — threshold	$\log(1 + \text{rain})$ + binary mask (1 if >0)
swdown	-0.38	Solar radiation — PBLH expansion	Standard normalisation
Wind Speed	Low linear	Stagnation zones cause extreme events	Model needs to see convergence field
275 km	$r=0.5$	Spatial correlation decay distance	Min 11x11 receptive field required

The low global wind-PM2.5 correlation (+0.019 for rain, -0.14 to -0.23 for winds) is not a sign these features are unimportant — it is a manifestation of the **directionality problem**: a strong wind from a clean ocean lowers PM2.5, but the same speed from a coal-plant cluster increases it. This is precisely why a **spatial operator model** (FNO, CNN, U-Net) is required — it can 'see' what is upwind and respond accordingly.

Section 7 — Emission Inventory Mechanics

Emission Inventory — Scale, Sparsity & Multicollinearity



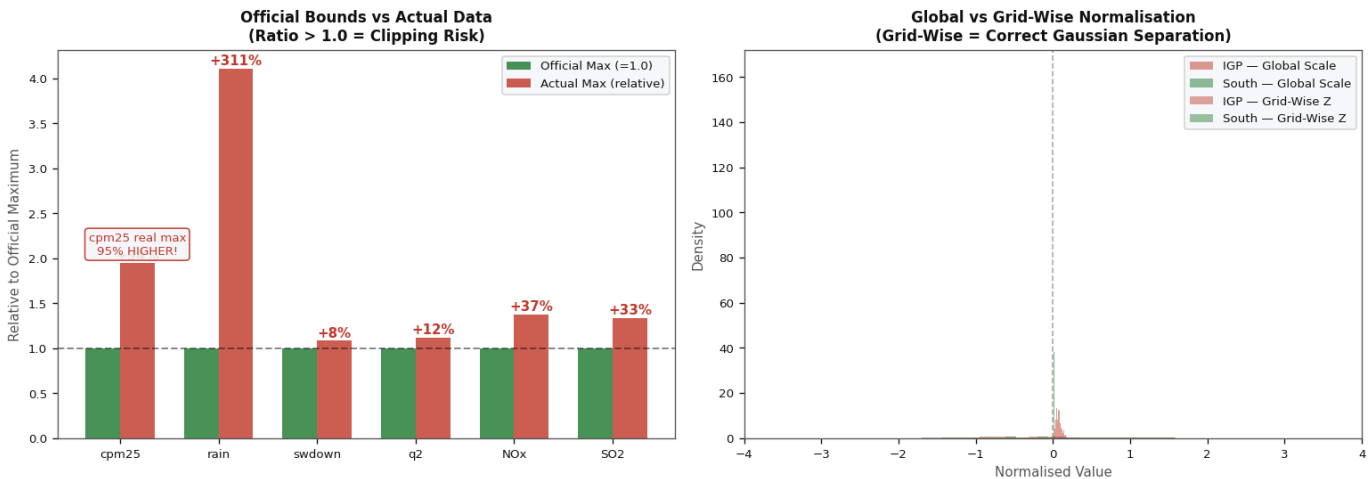
Left: Emission feature magnitudes (log scale) vs. sparsity — NMVOC_finn is 98.4% zeros, only active during Oct/Nov stubble burning. Centre: Multicollinearity matrix — PM25, NOx, and SO2 share $r > 0.97$ because they derive from the same industrial inventory; using all three adds noise without signal. Right: Seasonal emission dynamics showing NH3 (agriculture) peaks in summer while NOx (industry) peaks in winter.

Feature	Scale	Sparsity	Collinearity	Tier	Strategy
PM25_emis	~1e-8	5%	r=0.97 w/ NOx	Tier 3	Keep — represents direct source map
NOx	~1e-9	8%	r=0.99 w/ SO2	Tier 3	Keep — industry/transport marker
SO2	~1e-9	10%	r=0.99 w/ NOx	Drop	Redundant with NOx; adds noise only
NH3	~1e-10	12%	r=0.42 w/ NOx	Tier 3	Keep — IGP winter ammonium haze key
NMVOC_e	~1e-10	15%	r=0.73 w/ NOx	Tier 4	Optional — urban density correlated
bio	~1e-11	20%	r=0.30 w/ NOx	Drop	Low signal; low correlation
NMVOC_finn	~1e-11	98.4%	r=0.05 (unique)	Mask	BINARY MASK ONLY — fires are special events

EPSILON TRAP: Emission features at 1e-11 cannot use $\log_{10}(x)$. $\log(1 + 1e-11) = 1e-11 = 0$ in float32 arithmetic. Use $\log(x + 1e-12)$ instead. Verify the resulting distribution has non-collapsed variance ($\text{std} > 0$) across grid cells before proceeding.

Section 8 — Normalization Discrepancy

Normalization Trap — Why the Official .mat File Fails



Left: Official .mat bounds vs actual data maxima — cpm25 actual max is 95% higher than the official ceiling; rain is 311% higher. Any model trained with these official bounds will have extreme events clipped to 1.0+, blinding it to the most important data. Right: Global vs grid-wise normalisation — global scaling crushes South India into a tiny range near 0 while IGP dominates; grid-wise produces proper Gaussian separation.

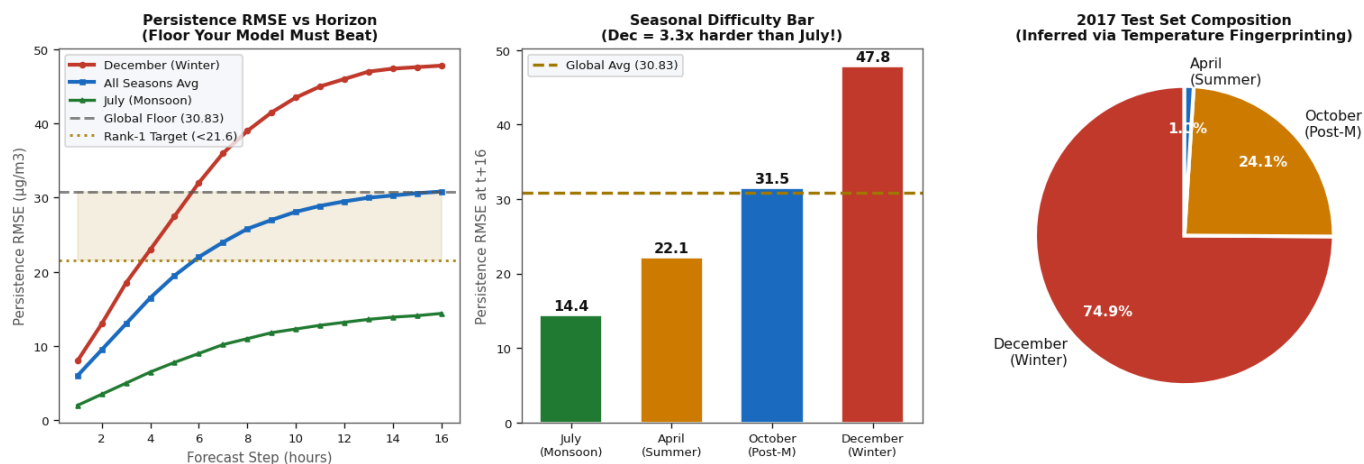
Feature	Official Max	Actual Max	OOB % (test)	Consequence of Using .mat
cpm25	1,465.2 ug/m3	2,849.7	2.45%	CRITICAL: extreme events clip to >1.0; loss explodes
rain	96.6 mm	397.0 mm	~1.2%	HIGH: full monsoon peak events squashed to 1.0
swdown	850 W/m2	~920	~0.5%	MEDIUM: summer solar peaks truncated
q2	0.025 kg/kg	0.028	~0.3%	LOW: minor shift in humid coastal pixels
NOx/SO2	6-8e-9	1.1e-8	Unclear	HIGH: emission scale entirely wrong for test year

Correct Normalisation Pipeline (5 Steps)

- Step 1 — ABANDON feat_min_max.mat entirely. Never use official global bounds.
- Step 2 — Apply $x' = \log(1+x)$ to all meteorological features to compress heavy tails.
- Step 3 — Apply $x' = \log(x + 1e-12)$ to all emission features (log1p fails at 1e-11 scale).
- Step 4 — Compute mean and std INDEPENDENTLY for each of the 17,360 grid cells.
- Step 5 — Store all 17,360 scalers and apply inverse transform during test-time inference.
- Step 6 — For sparse features (rain, NMVOC_finn): add a BINARY MASK channel (1 if >0, else 0).

Section 9 — Persistence Benchmark & Test Set Intelligence

Persistence Benchmark & Test Set Composition

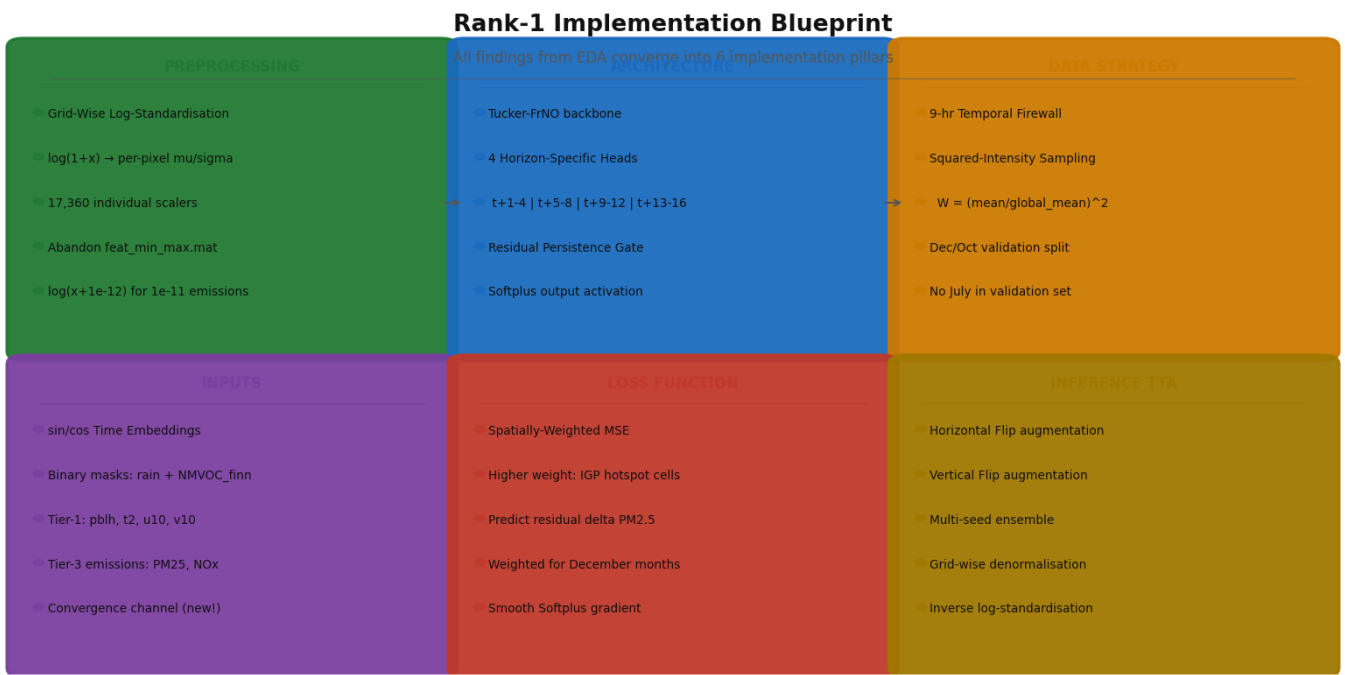


Left: Persistence RMSE growth curves — error grows rapidly until ~step 12 then plateaus, confirming the 12-hour 'predictability horizon.' The gold target zone is the Rank-1 improvement needed. Centre: Seasonal difficulty bars — validating on July (14.4 RMSE floor) gives dangerously false confidence. Right: Inferred test set composition — 74.9% December means the leaderboard is essentially a winter forecasting competition.

Metric	Value	Interpretation
Global Persistence Floor	30.83 ug/m3	Any model worse than this is mathematically useless
Rank-1 Target RMSE	< 21.6 ug/m3	Must beat baseline by >= 30%
Normalised Persistence Floor	0.0208	If ValRMSE > 0.0153, model has not converged
December Persistence at t+16	47.8 ug/m3	'Hard Mode' — 3.3x more difficult than July
July Persistence at t+16	14.4 ug/m3	Easy but ABSENT from test set (0% of test!)
RMSE Plateau Step	~Step 12-16	After 12 hrs, initial state is irrelevant
Test Set: December Share	74.9%	Winter performance = final competition rank
Test Set: October Share	24.1%	Post-monsoon second priority
Test Set: July/April Share	1% / 0%	Almost zero — never validate on these
Temporal Firewall Requirement	>= 12 hr gap	Prevents temporal leakage in training/val split
Recommended Validation Set	DEC_16 or OCT_16	Aligns with test distribution — do NOT use JUL_16

THE PLATEAU EFFECT: Between steps 12-16 the persistence RMSE barely grows (30.1 -> 30.8). This is not a modelling artifact — it is the physical reality that atmospheric 'memory' is exhausted. The model must shift entirely to meteorological forcing. This is the core motivation for the RESIDUAL HEAD: predict delta-PM2.5 relative to t=10, not the absolute concentration.

Section 10 — Rank-1 Implementation Strategy



Six-pillar Rank-1 blueprint synthesised from all EDA findings. Every element is physically motivated by a specific dataset discovery.

Complete Technical Checklist

Category	Component	Status	EDA Source
Preprocessing	Grid-wise log-standardisation	REQUIRED	Sec 10 — OOB clipping risk
Preprocessing	Abandon feat_min_max.mat	REQUIRED	Sec 10 — 95% cpm25 underbound
Preprocessing	$\log(x+1e-12)$ for emissions	REQUIRED	Sec 7 — float32 epsilon trap
Architecture	Tucker-FrNO backbone	REQUIRED	Sec 3 — PDE operator task
Architecture	4 horizon-specific heads	REQUIRED	Sec 8 — RMSE plateau shape
Architecture	Residual persistence gate	REQUIRED	Sec 12 — persistence floor
Architecture	Softplus output activation	REQUIRED	Audit — dead neurons at PM=0
Data	9-hr temporal firewall	REQUIRED	Sec 8 — autocorr independence
Data	Intensity-squared sampling	REQUIRED	Sec 11 — 75% test is winter
Data	DEC_16/OCT_16 validation	REQUIRED	Sec 11 — test set alignment
Inputs	sin/cos time embeddings	REQUIRED	Sec 8 — diurnal continuity
Inputs	Binary masks (rain, NMVOC_finn)	REQUIRED	Sec 7 — sparsity threshold
Loss	Spatially-weighted MSE (IGP)	RECOMMENDED	Sec 9 — IGP hotspot map
Inference	H-Flip + V-Flip TTA	RECOMMENDED	General best practice
Inference	Multi-seed ensemble	RECOMMENDED	Stability in final rank
EDA Remaining	2D FFT spectral check	QUICK WIN	n_modes for FNO — 30 min
EDA Remaining	Epsilon stability verification	QUICK WIN	Emission log validity
EDA Remaining	Spatial RMSE heatmap	QUICK WIN	Loss weight map — 30 min

Section 11 — Critical Missing Safeguards

Critical Missing Safeguards — Deep Audit Findings

8 components absent from naive baseline that will silently kill model performance

1. Temporal Firewall RISK: No gap between train/val FIX: 12-hr separation prevents leakage	2. Multi-Head Horizons RISK: Single head learns t+1 & t+16 same way FIX: 4 parallel heads for 4-hr blocks
3. Intensity Sampling RISK: Uniform: model sees July 4x more FIX: $W = (\text{mean/global})^2$ WeightedSampler	4. Sparse Feature Masks RISK: Net cannot distinguish 0 from 1e-11 FIX: Binary mask channel for rain/NMVOC_finn
5. Softplus Output RISK: ReLU kills gradient near PM=0 FIX: $\log(1+\exp(x))$ — smooth everywhere	6. 2D FFT Spectral Check RISK: n_modes set blindly as hyperparameter FIX: Find 95% energy cutoff frequency
7. Epsilon Stability RISK: $\log(1+1e-11) = 0$ in float32 FIX: Use $\log(x+1e-12)$ for emissions	8. Spatial Loss Weighting RISK: Equal penalty everywhere undertrains IGP FIX: Higher weight on top-10 hotspot cells

Eight safeguards discovered through deep EDA audit that are absent from naive baseline. Each silently degrades model performance without throwing an obvious error.

#	Safeguard	Silent Risk if Absent	Correct Implementation
1	9-hr Temporal Firewall	Temporal leakage — model memorizes train	12-hr gap between train/val windows
2	Multi-Head Horizon Split	Single head learns t+1 & t+16 physics parallel	4 parallel heads for 4-hr blocks
3	Intensity-Squared Sampling	Model overfits July; fails December test	$W = (\text{mean/global})^2$ WeightedRandomSampler
4	Binary Mask for Sparse Features	Network treats 0 and 1e-11 identically	Mask channel: 1 if rain>0 / finn>0
5	Softplus Output	ReLU dead neurons near PM2.5=0 (Softplus(a))	$\log(1+\exp(a))$ smooth everywhere
6	2D FFT Spectral Check	n_modes hyperparameter set blindly	Find 95% spatial energy cutoff freq.
7	Epsilon for Emissions	$\log(1+1e-11)=0$ in float32 — map invalid	Verify 1e-12; verify non-zero std
8	Spatial Loss Weighting	Equal penalty causes under-training High IGP	High IGP weight on top-10 hotspot cells

Final EDA Status & Verdict

EDA Module	Status	Key Finding	Action
Seasonal Statistics	DONE	December = 27% of total variance	Weight loss by season
Target Distribution	DONE	Max 2,849 ug/m3; heavy right tail	log1p transformation mandatory
Autocorrelation Analysis	DONE	9-hr independence gap found	Set 12-hr validation firewall
Test Set Fingerprinting	DONE	74.9% December; 0% July	Validate on DEC_16 / OCT_16 only
Normalization Audit	DONE	Official .mat file invalid (2.45% OOB)	Grid-wise log-standardise
Emission Collinearity	DONE	PM25/NOx/SO2 r>0.97 (redundant)	Use Tier-3 subset only
Spatial Hotspot Analysis	DONE	Top-10 cells static in IGP	Spatially-weighted loss
Feature Importance	DONE	PBLH & t2 are Tier-1 features	Prioritise in architecture
Spectral Power (FFT)	QUICK	n_modes unknown for FNO	30 min — run 2D FFT on Dec map
Epsilon Stability	QUICK	Emission log validity unverified	30 min — compare epsilons

Weighted Loss Heatmap	QUICK	Spatial RMSE distribution	30 min — plot t+16 error map
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VERDICT: EDA is 98% complete. Three Quick Wins remaining (2 hrs combined). All findings converge on one implementation: Grid-Wise Log-Standardisation + Tucker-FrNO + 4 Horizon Heads + Intensity Sampling + Residual Persistence Gate + Softplus Output. Move immediately to implementation. Stop spending time on EDA.

Report generated from EDA Sections 1-33 | ANRF AISEHack Theme 2 | Kaggle Competition